

Epidemics

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Ichiro Obara

UCLA

Econ142, Fall 2024

- We are going to study many types of diffusion on networks
- Here we study various epidemics models, where each person is infected by a disease/virus through **contact network**.
- First we present some classical models of epidemics without explicit network structures.

Compartment Models

- There is a family of models called **compartment models**, which have been used in epidemiology from the early 20th century. In these models, the population is divided into a few compartments based on the stage of infection.
 - ▶ **Susceptible:** People in this category are susceptible to infection from their contacts.
 - ▶ **Infected:** People in this category has caught the disease and can infect their contacts.
 - ▶ **Removed:** People in this category has caught the disease but cannot infect their contacts any more (they have recovered with full immunity or died).

Some Basic Models

- **SI model:** People are initially susceptible, then may be infected by their contacts. Once infected, they stay infectious forever.
- **SIR model:** People are initially susceptible, then may be infected by their contacts. After infected, they will be removed at some point.
- **SIS model:** People are initially susceptible, then may be infected by their contacts. After infected, they will recover without immunity and become susceptible again.
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SI Model

- Let $S(t)$ be the number susceptible people and $I(t)$ be the number of infected people at period t . The total number of people is always $N = S(t) + I(t)$.

- Let β be the average number of contacts per unit of time.

- $S(t)$ and $I(t)$ follow the following dynamics:

$$S(t+1) - S(t) = -\frac{\beta S(t)I(t)}{N}, \quad I(t+1) - I(t) = \frac{\beta S(t)I(t)}{N}$$

- There is a random contact network in the background.

SI Model

- In terms of the ratios $s(t) = \frac{S(t)}{N}$ and $i(t) = \frac{I(t)}{N}$,

$$s(t+1) - s(t) = -\beta s(t)i(t), \quad i(t+1) - i(t) = \beta s(t)i(t)$$

- Since $s(t) + i(t) = 1$, one of the equations is redundant. Drop the first equation and replace $s(t)$ with $1 - i(t)$. Making the time interval short, we obtain the following differential equation for $i(t)$.

$$\frac{di(t)}{dt} = \beta(1 - i(t))i(t)$$

- The solution is $i(t) = \frac{i(0)e^{\beta t}}{1 - i(0) + i(0)e^{\beta t}}$, where $i(0)$ is the initial seed of disease. This is a logistic function. The infection rate increases monotonically (fastest at $i = 0.5$) and converges to 1 independent of the size of $i(0) > 0$.

SIR Model

- For SIR model, the infected people is removed and become noninfectious. Let $R(t)$ be the number of removed people at period t and γ be the rate of removal.

The dynamics of this model is described by

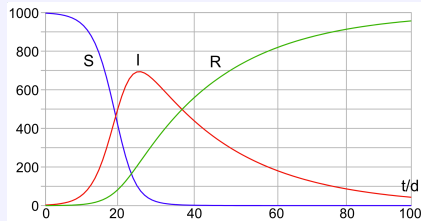
$$\begin{aligned} S(t+1) - S(t) &= -\frac{\beta S(t)I(t)}{N} \\ I(t+1) - I(t) &= \frac{\beta S(t)I(t)}{N} - \gamma I(t) \\ R(t+1) - R(t) &= \gamma I(t) \end{aligned}$$

- Using ratios and continuous time,

$$\begin{aligned} \frac{ds(t)}{dt} &= -\beta s(t)i(t) \\ \frac{di(t)}{dt} &= \beta s(t)i(t) - \gamma i(t) \\ \frac{dr(t)}{dt} &= \gamma i(t) \end{aligned}$$

- Eliminate i from the differential equation for $s(t)$ and $r(t)$. Then we have $\frac{1}{s(t)} \frac{ds(t)}{dt} = -\frac{\beta}{\gamma} \frac{dr(t)}{dt}$
- Integrating this and assuming $r(0) = 0$, $s(t) = s(0)e^{-\frac{\beta}{\gamma}r(t)}$. If we plug this into the differential equation for $r(t)$ using $s(t) + r(t) + i(t) = 1$, we get $\frac{dr(t)}{dt} = \gamma(1 - s(0)e^{-\frac{\beta}{\gamma}r(t)} - r(t))$.
- Once we solve this to get $r(t)$, then we can derive the rest: $s(t)$ and $i(t)$.

- $s(t)$ is decreasing and $r(t)$ is increasing over time.
- An example of dynamics from Wikipedia
($S(0) = 997, I(0) = 3, R(0) = 0, \beta = 4, \gamma = 0.04$).



- We can find the fraction of infected/uninfected people in the long run by using

$$\frac{dr(t)}{dt} = \frac{ds(t)}{dt} = 0 \text{ and } s(\infty) \approx 1:$$

$$\begin{aligned} \blacktriangleright r(\infty) &= 1 - e^{-\frac{\beta}{\gamma}r(\infty)} \\ \blacktriangleright s(\infty) &= e^{-\frac{\beta}{\gamma}r(\infty)}. \end{aligned}$$

- Note that $r(\infty) + s(\infty) = 1$ and $r(\infty) > 0 \Leftrightarrow \frac{\beta}{\gamma} > 1$ (looks familiar?)

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Basic Reproduction Number

- **Basic reproduction number** is the expected number of people who are infected by one infected individual (before becoming noninfectious) in the initial phase of epidemics ($i(t) \approx 0$)
- In SIR model, the basic reproduction number is $\frac{\beta}{\gamma}$.
- So, the previous result shows that a positive fraction of population is eventually infected when the basic reproduction number is more than 1.

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SIS Model

- For SIS model, the infected people recover and become susceptible again.

Let γ be the rate of recovery this time. The dynamical system is given by

$$S(t+1) - S(t) = -\frac{\beta S(t)I(t)}{N} + \gamma I(t), \quad I(t+1) - I(t) = \frac{\beta S(t)I(t)}{N} - \gamma I(t)$$

- Again using the ratios and dropping the first equation, we have the following differential equation.

$$\frac{di(t)}{dt} = (\beta - \gamma - \beta i(t))i(t)$$

- The solution is $i(t) = (1 - \frac{\gamma}{\beta}) \frac{Ce^{(\beta-\gamma)t}}{1 + Ce^{(\beta-\gamma)t}}$

where C is determined by the initial condition $i(0) = (1 - \frac{\gamma}{\beta}) \frac{C}{1+C}$.

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SIS Model

- $\frac{\beta}{\gamma}$ is another instance of **basic reproduction number**. When it is larger than 1, the (small) infected cluster expands in size. When it is smaller than one, the infected cluster shrinks in size.
- When $\beta > \gamma$, $i(t)$ is a logistic curve like SI model. But it converges to $1 - \frac{\gamma}{\beta}$ instead of 1, where $\frac{di(t)}{dt} = 0$ as the inflow and the outflow of infection balances.
- When $\beta < \gamma$, the fraction of infected population converges to 0.

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Random Network as Contact Network for Epidemics

- Consider the following dynamic and stochastic model of epidemics.
 - n nodes.
 - Some nodes are infected initially. Each infected node contacts and infects other $n - 1$ nodes with probability p independently. So, the expected basic reproduction number is $(n - 1)p$.
- Note that the infected population at the end of the process corresponds to the component of random graphs $G(n, p)$ that contain some initially infected seeds.
- Remember that there is a unique giant component if and only if $p > \frac{1}{n}$ as $n \rightarrow \infty$. For large n , this is essentially the same as the average infection rate $(n - 1)p$ being more than 1.

Long-Run Infection Rate with Intervention

- Suppose that, before the disease starts spreading, we vaccinate π fraction of population, who then become **immune** to the disease.
- In terms of random graph, this corresponds to $G(n(1 - \pi), p)$ instead of $G(n, p)$.
- Then the condition for a giant component becomes $p \leq \frac{1}{n(1 - \pi)}$. This is as if the infection rate p is reduced to $p(1 - \pi)$ for a random graph with n nodes ($G(n, p(1 - \pi))$).
- Suppose that $p > \frac{1}{n}$, but $p(1 - \pi) < \frac{1}{n}$. Then the infected people would occupy a positive share of the population without vaccination, but they would become negligible (relative to the whole population) with vaccination.

- We can consider different kinds of interventions and their impact on the eventual infection rate.
- Let $P(\cdot)$ be a degree distribution that follows from some intervention.
- Let s be the size of the (unique) giant component when n is large. The following equation is useful to determine s . $1 - s = \sum_{d=0}^{n-1} P(d)(1 - s)^d$
 - LHS is the probability that a node is outside of the giant component.
 - RHS is the probability that every neighbor of the node is outside of the giant component.
- For $G(n, p)$ with Poisson degree distribution (as $n \rightarrow \infty$), the RHS becomes $e^{-\lambda s}$, where $\lambda = (n - 1)p$. So this is consistent with our previous formula for the size of giant component.

- The previous uniform vaccination intervention generates the degree distribution $P_\pi(0) = \pi + (1 - \pi)P(0)$ and $P_\pi(d) = (1 - \pi)P(d)$. Hence the RHS becomes $\pi + (1 - \pi)e^{-\lambda s}$. So, $s = (1 - \pi)(1 - e^{-\lambda s})$.
- Consider another intervention that prevents more than 10 contacts. This corresponds to a static random graph where the degree distribution is approximately Poisson up to $d = 9$, then every $d \geq 10$ is counted as $d = 10$ (hence $P_{10}(d) = P(d)$ for $d < 10$ and $P_{10}(10) = \sum_{d=10}^{\infty} P(d)$).
- This intervention reduces the RHS of the giant component formula, hence reduces the size of the giant component.

Summary

- Various basic compartment models and their dynamics.
- Random network as contact network for epidemics.
- Long-run share of infected people = a giant component of a random network. Study the effect of interventions through their impact on the degree distribution.

Suggested reading. EK Ch.21.1 - Ch.21.4